

Лабораторная работа №13

Статическая маршрутизация в Интернете. Планирование

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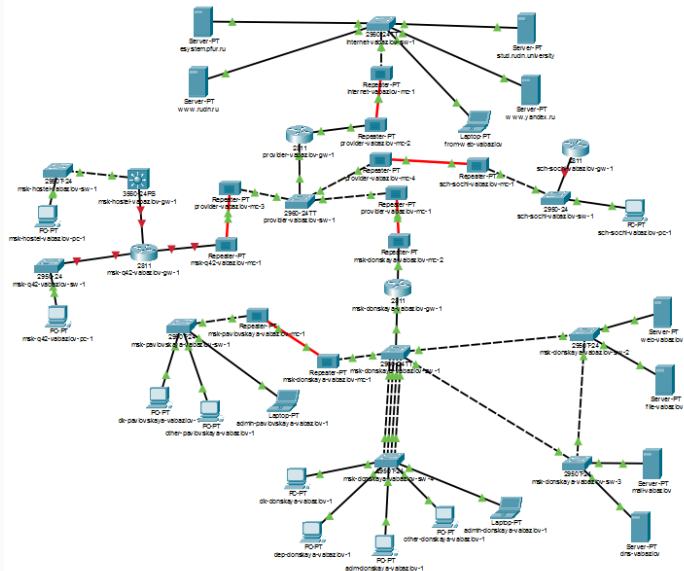
Цель работы

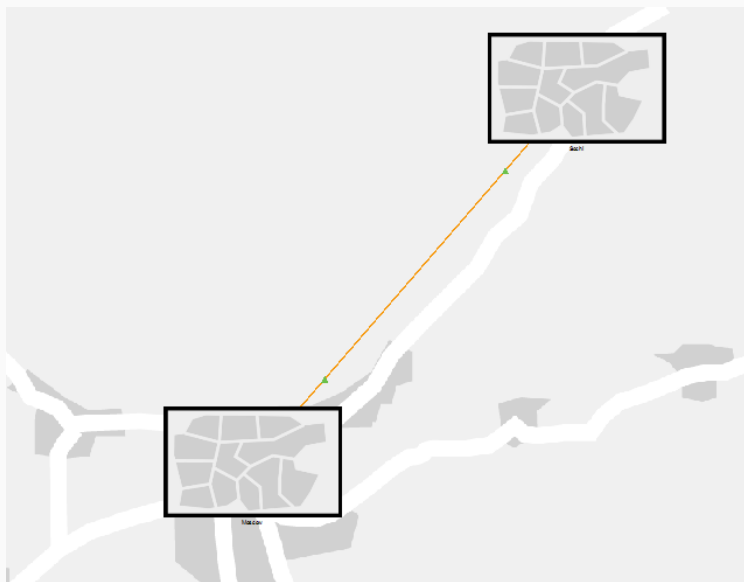
Провести подготовительные мероприятия по организации взаимодействия через сеть провайдера посредством статической маршрутизации.

Необходимо связать локальную сеть с:

- сетью основной территории организации в 42-м квартале в Москве;
- сетью филиала в г. Сочи.

Дополнение схемы проекта





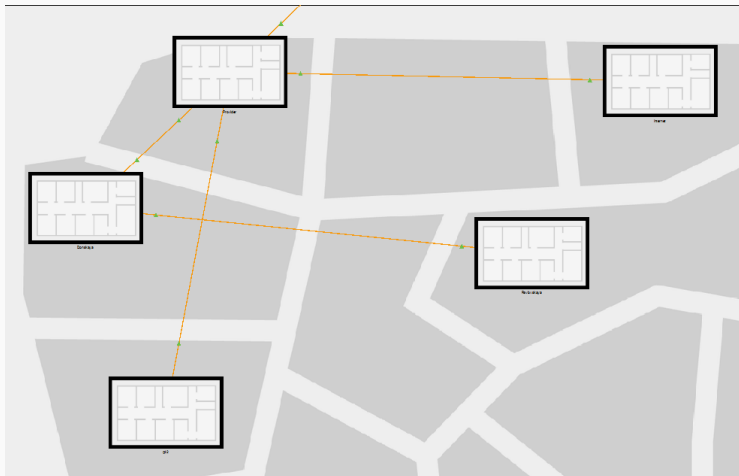
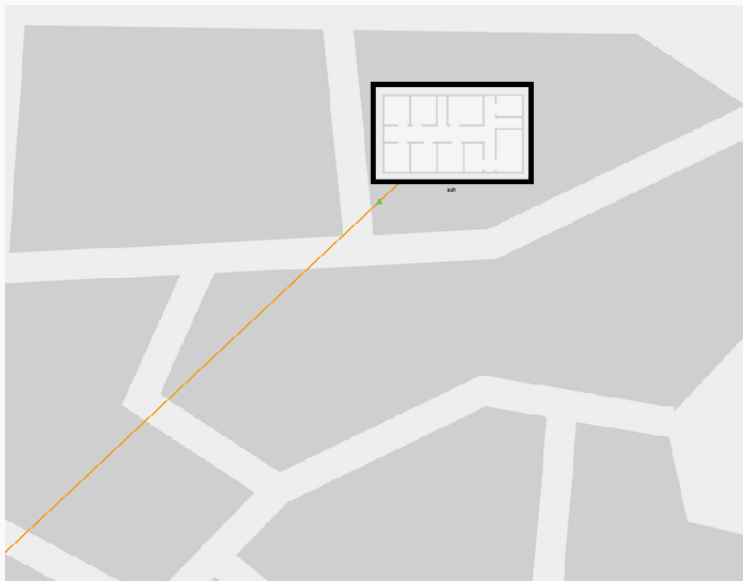


Рис. 3: Схема размещения узлов



Первоначальная настройка оборудования

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host msk-q42-vabazlov-gw-1
msk-q42-vabazlov-gw-1(config)#line vty 0 4
msk-q42-vabazlov-gw-1(config-line)#pass cisco
msk-q42-vabazlov-gw-1(config-line)#login
msk-q42-vabazlov-gw-1(config-line)#line console 0
^
% Invalid input detected at '^' marker.

msk-q42-vabazlov-gw-1(config-line)#pass cisco
msk-q42-vabazlov-gw-1(config-line)#line console 0
msk-q42-vabazlov-gw-1(config-line)#pass cisco
msk-q42-vabazlov-gw-1(config-line)#login
msk-q42-vabazlov-gw-1(config-line)#exit
msk-q42-vabazlov-gw-1(config)#enab sec cisco
msk-q42-vabazlov-gw-1(config)#serv pass
msk-q42-vabazlov-gw-1(config)#
msk-q42-vabazlov-gw-1(config)#user admin priv 1 sec cisco
msk-q42-vabazlov-gw-1(config)#ip domain-name q42.rudn.edu
msk-q42-vabazlov-gw-1(config)#cryp key gen rsa
The name for the keys will be: msk-q42-vabazlov-gw-1.q42.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-q42-vabazlov-gw-1(config)#line vty 0 4
*Mar 1 0:12:36.375: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:12:36.375: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-q42-vabazlov-gw-1(config-line)#trans in ssh
msk-q42-vabazlov-gw-1(config-line)#^Z
msk-q42-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-q42-vabazlov-gw-1#
```

Коммутатор 42-го квартала

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host msk-q42-vabazlov-sw-1
msk-q42-vabazlov-sw-1(config)#line vty 0 4
msk-q42-vabazlov-sw-1(config-line)#pass cisco
msk-q42-vabazlov-sw-1(config-line)#login
msk-q42-vabazlov-sw-1(config-line)#line cons 0
msk-q42-vabazlov-sw-1(config-line)#pass cisco
msk-q42-vabazlov-sw-1(config-line)#login
msk-q42-vabazlov-sw-1(config-line)#exit
msk-q42-vabazlov-sw-1(config)#enab sec cisco
msk-q42-vabazlov-sw-1(config)#serv pass
msk-q42-vabazlov-sw-1(config)#user admin priv 1 sec cisco
msk-q42-vabazlov-sw-1(config)#ip domain-name q42.rudn.edu
msk-q42-vabazlov-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-q42-vabazlov-sw-1.q42.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-q42-vabazlov-sw-1(config)#line vty 0 4
*Mar 1 0:14:16.714: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:14:16.714: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-q42-vabazlov-sw-1(config-line)#trans in ssh
msk-q42-vabazlov-sw-1(config-line)#^Z
msk-q42-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-vabazlov-sw-1#wr mem
Building configuration...
[OK]
msk-q42-vabazlov-sw-1#
```

Оборудование сегмента Hostel

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-hostel-vabazlov-gw-1
msk-hostel-vabazlov-gw-1(config)#line vty 0 4
msk-hostel-vabazlov-gw-1(config-line)#pass cisco
msk-hostel-vabazlov-gw-1(config-line)#login
msk-hostel-vabazlov-gw-1(config-line)#line cons 0
msk-hostel-vabazlov-gw-1(config-line)#pass cisco
msk-hostel-vabazlov-gw-1(config-line)#login
msk-hostel-vabazlov-gw-1(config-line)#exit
msk-hostel-vabazlov-gw-1(config)#enab sec cisco
msk-hostel-vabazlov-gw-1(config)#serv pass
msk-hostel-vabazlov-gw-1(config)#user admin priv 1 sec cisco
msk-hostel-vabazlov-gw-1(config)#ip ssh ver 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
msk-hostel-vabazlov-gw-1(config)#ip domain-name hostel.rudn.rdu
msk-hostel-vabazlov-gw-1(config)#cryp key gen rsa
The name for the keys will be: msk-hostel-vabazlov-gw-1.hostel.rudn.rdu
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-hostel-vabazlov-gw-1(config)#line vty 0 4
*Mar 1 0:16:13.314: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:16:13.314: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-hostel-vabazlov-gw-1(config-line)#trans in ssh
msk-hostel-vabazlov-gw-1(config-line)#^Z
msk-hostel-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-vabazlov-gw-1#
```

Коммутатор сегмента Hostel

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-hostel-vabazlov-sw-1
msk-hostel-vabazlov-sw-1(config)#line vty 0 4
msk-hostel-vabazlov-sw-1(config-line)#pass cisco
msk-hostel-vabazlov-sw-1(config-line)#login
msk-hostel-vabazlov-sw-1(config-line)#line cons 0
msk-hostel-vabazlov-sw-1(config-line)#pass cisco
msk-hostel-vabazlov-sw-1(config-line)#login
msk-hostel-vabazlov-sw-1(config-line)#exit
msk-hostel-vabazlov-sw-1(config)#enab sec cisco
msk-hostel-vabazlov-sw-1(config)#serv pass
msk-hostel-vabazlov-sw-1(config)#user admin priv 1 sec cisco
msk-hostel-vabazlov-sw-1(config)#ip domain-name hostel.rudn.edu
msk-hostel-vabazlov-sw-1(config)#crypt key gen rsa
The name for the keys will be: msk-hostel-vabazlov-sw-1.hostel.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-hostel-vabazlov-sw-1(config)#line vty 0 4
*Mar 1 0:18:18.602: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:18:18.602: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-hostel-vabazlov-sw-1(config-line)#trans in ssh
msk-hostel-vabazlov-sw-1(config-line)#^Z
msk-hostel-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-vabazlov-sw-1#wr mem
Building configuration...
[OK]
msk-hostel-vabazlov-sw-1#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#line vty 0 4
Switch(config-line)#pass cisco
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#line cons 0
Switch(config-line)#pass cisco
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#enab sec cisco
Switch(config)#serv pass
Switch(config)#user admin priv 1 sec cisco
Switch(config)#ip domain-name sochi.rudn.edu
Switch(config)#crypto key gen rsa
% Please define a hostname other than Switch.
Switch(config)#host sch-sochi-vabazlov-sw-1
sch-sochi-vabazlov-sw-1(config)#crypto key gen rsa
The name for the keys will be: sch-sochi-vabazlov-sw-1.sochi.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

sch-sochi-vabazlov-sw-1(config)#line vty 0 4
*Mar 1 0:16:37.970: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:16:37.970: %SSH-5-ENABLED: SSH 1.5 has been enabled
sch-sochi-vabazlov-sw-1(config-line)#trans in ssh
sch-sochi-vabazlov-sw-1(config-line)#^Z
sch-sochi-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-sw-1#
```

Маршрутизатор филиала в Сочи

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hos
Router(config)#hostname sch-sochi-vabazlov-gw-1
sch-sochi-vabazlov-gw-1(config)#line vty 0 4
sch-sochi-vabazlov-gw-1(config-line)#pass cisco
sch-sochi-vabazlov-gw-1(config-line)#login
sch-sochi-vabazlov-gw-1(config-line)#exit
sch-sochi-vabazlov-gw-1(config)#line console 0
sch-sochi-vabazlov-gw-1(config-line)#pass cisco
sch-sochi-vabazlov-gw-1(config-line)#login
sch-sochi-vabazlov-gw-1(config-line)#exit
sch-sochi-vabazlov-gw-1(config)#enab sec cisco
sch-sochi-vabazlov-gw-1(config)#serv pass
sch-sochi-vabazlov-gw-1(config)#user admin priv 1 sec cisco
sch-sochi-vabazlov-gw-1(config)#ip domain-name sochi.rudn.edu
sch-sochi-vabazlov-gw-1(config)#crypto key gen rsa
The name for the keys will be: sch-sochi-vabazlov-gw-1.sochi.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

sch-sochi-vabazlov-gw-1(config)#line vty 0 4
*Mar 1 0:18:44.668: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:18:44.668: %SSH-5-ENABLED: SSH 1.5 has been enabled
sch-sochi-vabazlov-gw-1(config-line)#trans in ssh
sch-sochi-vabazlov-gw-1(config-line)#^Z
sch-sochi-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-vabazlov-gw-1#
```

Схемы L1, L2 и L3

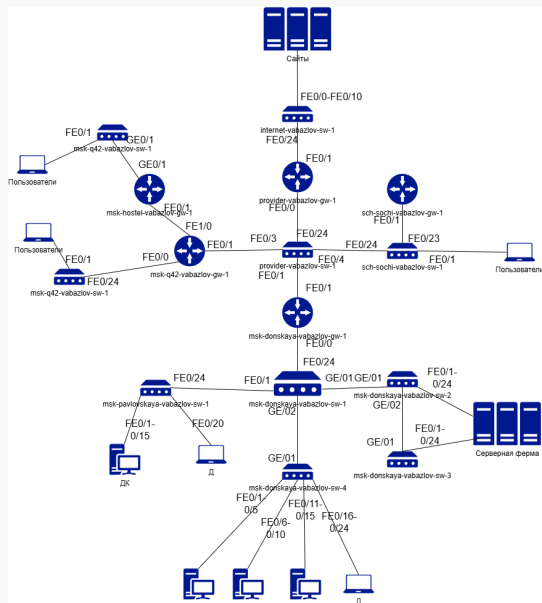
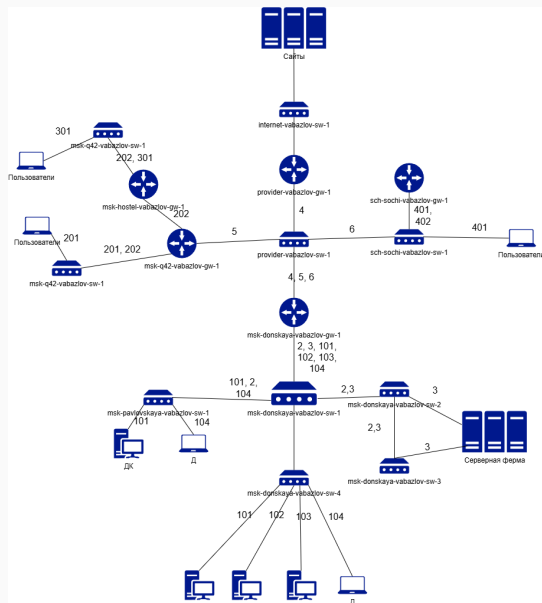
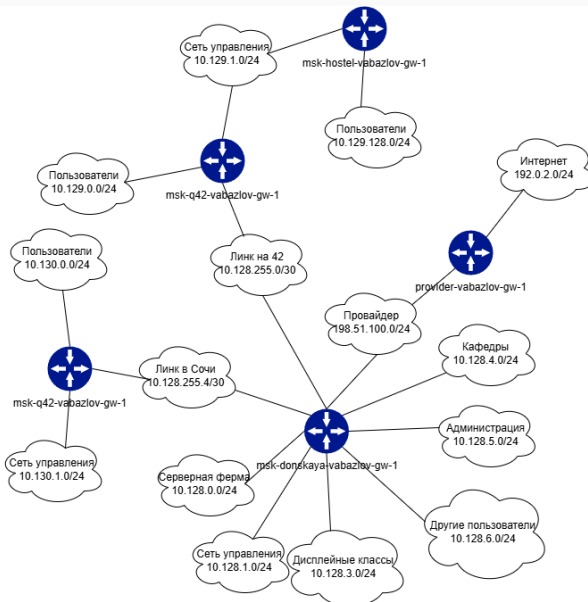


Схема L2





Итоги работы

В ходе лабораторной работы:

- дополнена схема проекта в Cisco Packet Tracer;
- добавлены сети 42-го квартала в Москве и филиала в Сочи;
- выполнена первоначальная настройка маршрутизаторов и коммутаторов;
- настроены параметры доступа и SSH;
- подготовлены схемы L1, L2 и L3;
- рассмотрены принципы статической маршрутизации.

Полученные результаты подготовили проект к дальнейшей настройке маршрутизации между подсетями и VLAN.